

- 5 (a) Simple harmonic motion is defined as the motion of a particle about a fixed point such that its acceleration is proportional to its displacement from the fixed point and is directed towards the point (or opposite in direction to the displacement). [B1]

(b) $p = mv$ ----- (1)

$E_K = \frac{1}{2}mv^2$ ----- (2) [C1 for both (1) & (2)]

sub (1) into (2)

$$E_K = \frac{p^2}{2m}$$

$$m = \frac{p^2}{2E_K}$$

$$= \frac{0.72^2}{2(0.86)} \quad [M1]$$

$$= 0.30 \text{ kg} \quad [A0]$$

(c) (i) $T_{\text{rod}} = T_{\text{pendulum}}$

$$\frac{1}{0.55} = 2\pi\sqrt{\frac{L}{9.81}} \quad [M1]$$

$$L = 0.82 \text{ m} \quad [A1]$$

(ii) $\Delta\phi = \frac{\Delta t}{T}(2\pi)$

$$= \frac{0.50}{1.82}(2\pi)$$

$$= 1.73 \text{ rad} \quad [B1]$$

- (d) (i) Damping due to viscous forces [B1]

(ii)

$$E_{\text{total}} = \frac{1}{2}m\omega^2 x_o^2$$

$$= \frac{1}{2}m\left(\frac{2\pi}{T}\right)^2 x_o^2$$

$$E_f - E_i = \frac{1}{2}m\left(\frac{2\pi}{T}\right)^2 (x_{o_f}^2 - x_{o_i}^2)$$

$$= \frac{1}{2}(0.30)\left(\frac{2\pi}{1.05 \times 2}\right)^2 (0.016^2 - 0.020^2) \quad [M1]$$

$$= -1.93 \times 10^{-4} \text{ J}$$

decrease in $E = 1.93 \times 10^{-4} \text{ J} \quad [A1]$

